Customer Churn Prediction - API Deployment Proof

1. Deployment Architecture

Stack Used

- FastAPI (Inference Layer)
- Scikit-learn (Logistic Regression)
- Joblib (Model Persistence)
- Docker (Containerization)
- Uvicorn (ASGI Server)

Container Command Used

```
docker run -p 8000:8000 churn-api
```

2. API Endpoint Test (Swagger UI Screenshot)

API Test Details

- Endpoint: POST /predict
- Input: Feature vector (preprocessed customer attributes)
- Decision Threshold: 0.4
- Output:
    - prediction: 0
    - churn_probability: 0.0072

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- HTTP Status: 200 OK

3. cURL Verification

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "features": [1,0,45,2,120,1,0,3,60,0,1,1,0,2,5,10,3,4,1,0,0,1,2,3,4,5,6,7,8,9]
}'
```

Request URL

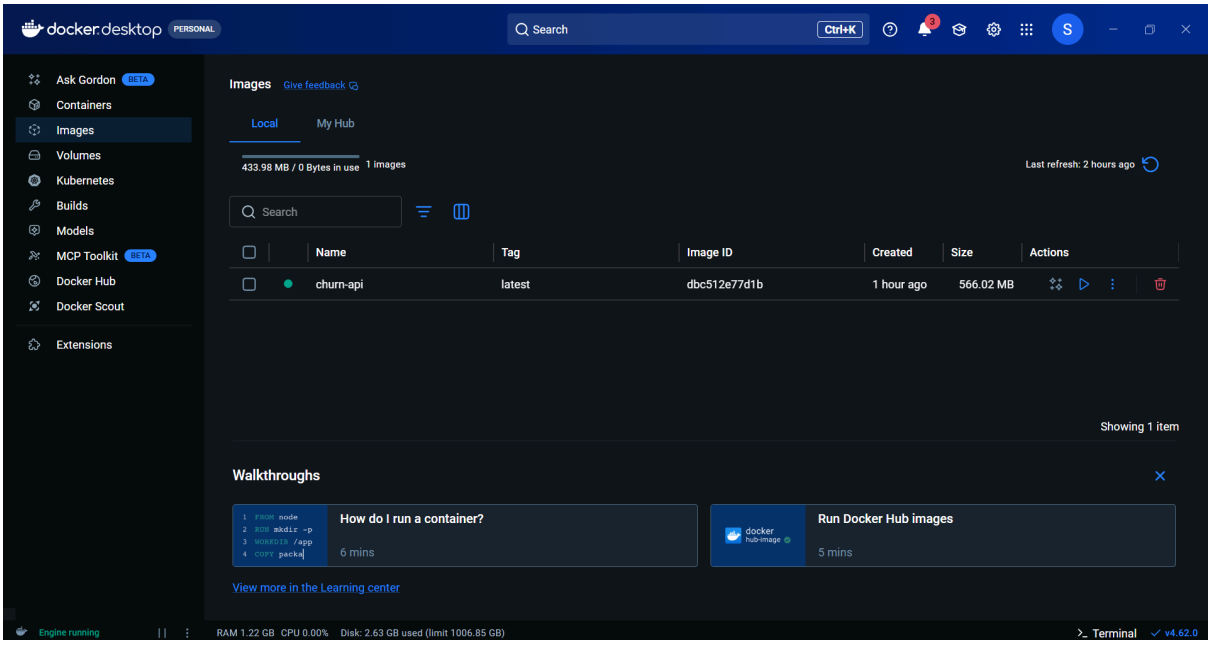
```
http://localhost:8000/predict
```

Server response

This shows:

- It works outside Swagger
- It's a real API endpoint

4. Deployment Confirmation



- Running container
- Port mapping 8000:8000
- Image name churn-ap

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5. Business Interpretation

- The deployed API serves real-time churn predictions using a tuned Logistic Regression model (ROC-AUC: 0.835).
- Customers scoring above a probability threshold of 0.4 are flagged as high churn risk.
- This enables integration into CRM systems for proactive retention campaigns.